114-1 Robot Perception and Learning

How to design an academic poster

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Poster Design Principles and Presentation Guidelines

- **Visuals over text:** Prioritize charts and figures, using text only to support and clarify key ideas.
- **Effective layout:** Place the most important content at the center or in visually prominent areas.
- **Use whitespace to guide attention:** Leverage whitespace to separate sections and help the audience quickly identify topics.
- **Enhance your presentation with live demos:** Bring a laptop or iPad for on-site demonstrations to showcase your results.

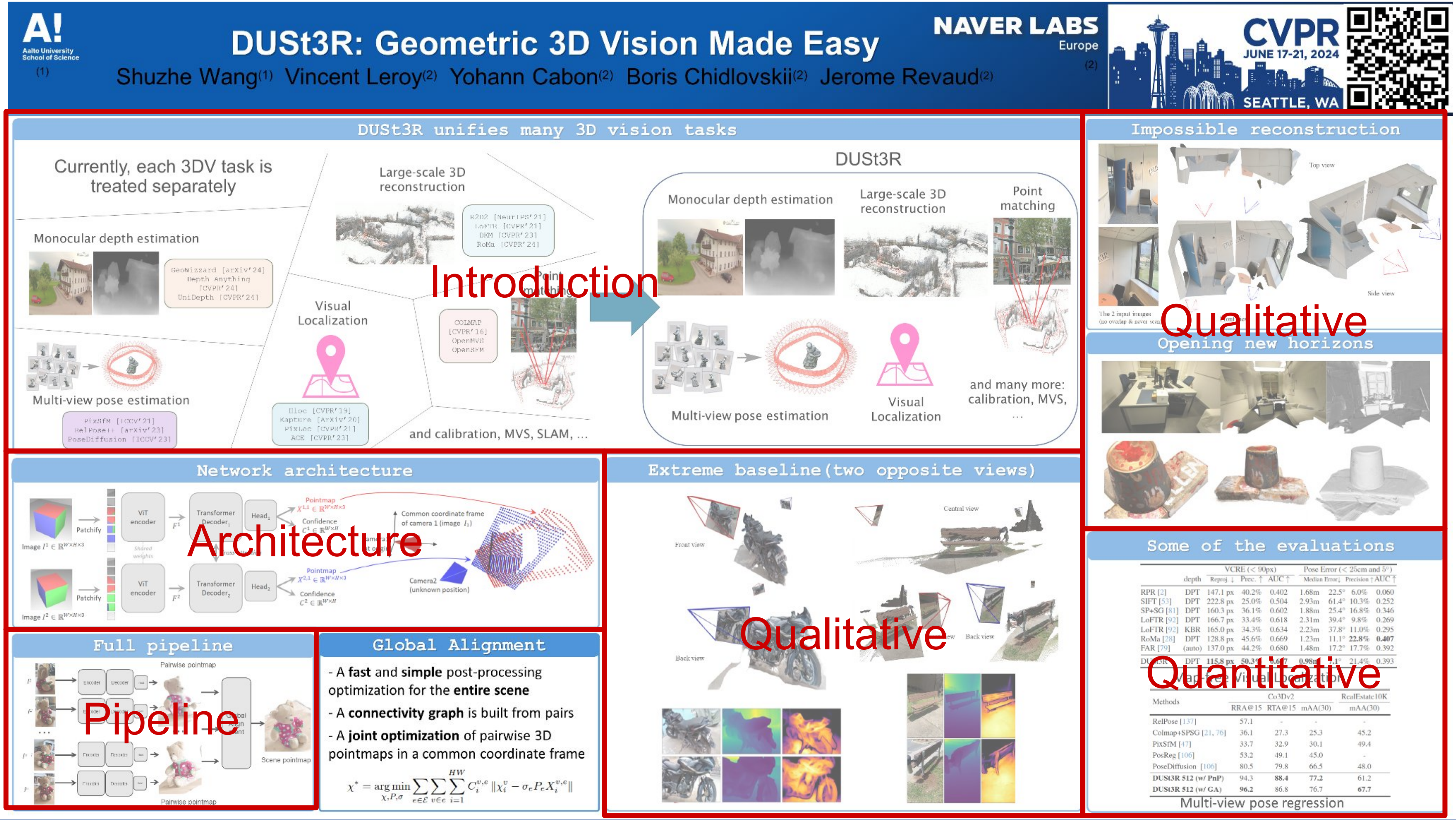
Considering that many of you are creating a poster for the first time, the TA has prepared a **poster template** for your reference. However, please feel free to express your own creativity — the TA does not want every group's poster to look the same.

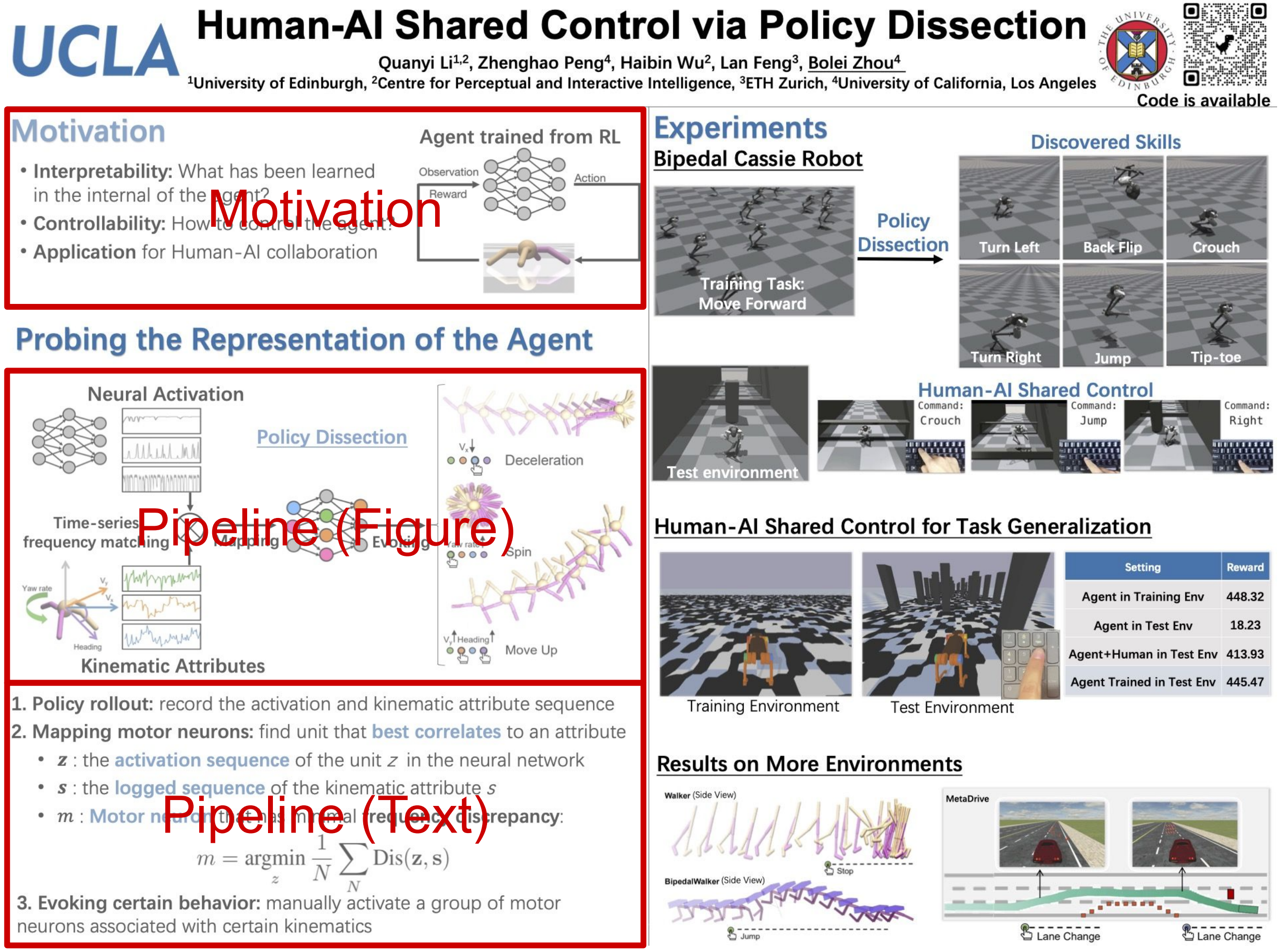
Formatting Guidelines

- A0 size, portrait orientation
- File name: EV_groupX.pdf
- Poster title: 72 pt
- Section titles: 48 - 60 pt
- Body text: 34 - 40 pt

Poster Layout Design

1. Title, Authors, Institution (logo), QR Code (project page)
2. Introduction / Motivation / Problem Formulation / Challenge
3. Key Insights / Takeaways
4. Methods
    - Pipeline, Architecture,…
5. Experiment Results
    - Quantitative, Qualitative, Ablation Study


Aalto University School of Science
DUSt3R: Geometric 3D Vision Made Easy
Shuzhe Wang(1) Vincent Leroy(2) Yohann Cabon(2) Boris Chidlovskii(2) Jerome Revaud(2)
NAVER LABS Europe
CVPR JUNE 17-21, 2024 SEATTLE, WA
DUSt3R unifies many 3D vision tasks
Currently, each 3DV task is treated separately
Large-scale 3D reconstruction
Monocular depth estimation
Visual Localization
Multi-view pose estimation
and calibration, MVS, SLAM, ...
DUSt3R
Point matching
and many more: calibration, MVS, ...
Introduction
Impossible reconstruction
Top view
Side view
Qualitative
Opening new horizons
Network architecture
Patchify
ViT encoder
Transformer Decoder
Shared weights
Pointmap
Confidence
Common coordinate frame of camera 1
Camera2 (unknown position)
Architecture
Extreme baseline (two opposite views)
Front view
Central view
Back view
Qualitative
Some of the evaluations
Quantitative
Multi-view pose regression
Full pipeline
Pairwise pointmap
Scene pointmap
Pipeline
Global Alignment
- A fast and simple post-processing optimization for the entire scene
- A connectivity graph is built from pairs
- A joint optimization of pairwise 3D pointmaps in a common coordinate frame


UCLA
Human-AI Shared Control via Policy Dissection
Quanyi Li[1,2], Zhenghao Peng[4], Haibin Wu[2], Lan Feng[3], Bolei Zhou[4]
[1]University of Edinburgh, [2]Centre for Perceptual and Interactive Intelligence, [3]ETH Zurich, [4]University of California, Los Angeles
Code is available
Motivation
• Interpretability: What has been learned in the internal of the agent?
• Controllability: How to control the agent?
• Application for Human-AI collaboration
Agent trained from RL
Observation
Reward
Action
Motivation
Probing the Representation of the Agent
Neural Activation
Policy Dissection
Time-series frequency matching
Mapping
Evoking
Deceleration
Spin
Move Up
Yaw rate
Heading
Kinematic Attributes
Pipeline (Figure)
1. Policy rollout: record the activation and kinematic attribute sequence
2. Mapping motor neurons: find unit that best correlates to an attribute
• z : the activation sequence of the unit z in the neural network
• s : the logged sequence of the kinematic attribute s
• m : Motor neuron that has minimal frequency discrepancy:
$m = \operatorname{argmin}_z \frac{1}{N} \sum_N \mathrm{Dis}(z, s)$
3. Evoking certain behavior: manually activate a group of motor neurons associated with certain kinematics
Pipeline (Text)
Experiments
Bipedal Cassie Robot
Training Task: Move Forward
Policy Dissection
Discovered Skills
Turn Left
Back Flip
Crouch
Turn Right
Jump
Tip-toe
Test environment
Human-AI Shared Control
Command: Crouch
Command: Jump
Command: Right
Human-AI Shared Control for Task Generalization
Training Environment
Test Environment
Setting | Reward
Agent in Training Env | 448.32
Agent in Test Env | 18.23
Agent+Human in Test Env | 413.93
Agent Trained in Test Env | 445.47
Results on More Environments
Walker (Side View)
Stop
BipedalWalker (Side View)
Jump
MetaDrive
Lane Change
Lane Change

Planning-oriented Autonomous Driving

Yihan Hu*, Jiazhi Yang*, Li Chen*, Keyu Li*, Chonghao Sima, Xizhou Zhu, Siqi Chai, Senyao Du, Tianwei Lin, Wenhai Wang, Lewei Lu, Xiaosong Jia, Qiang Liu, Jifeng Dai, Yu Qiao, Hongyang Li

上海人工智能实验室 Shanghai Artificial Intelligence Laboratory · 商汤 sensetime

Award Candidate

JUNE 18-22, 2023 CVPR VANCOUVER, CANADA

Summary

Summary or Takeaways

- **Planning-oriented Philosophy**: An end-to-end autonomous driving framework in pursuit of safe-planning, facilitated with various well-organized AD tasks.
- **Unified Query design**: Queries as interfaces to coordinate all tasks in training and transmit upstream knowledge to planner.
- **SOTA Performance** with vision-only input: SOTAs on all tasks, even surpassing LiDAR-based framework on planning.

Motivation: Towards Reliable Planning

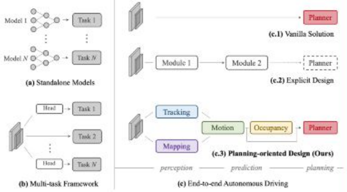
Limitations of prior AD frameworks:

- **Standalone Models:**
  Error accumulation; feature misalignment across modules
- **Multi-task Framework:**
  Inefficient tasks' coordination or knowledge transferring
- **End-to-end Framework:**
  (1) Vanilla: Direct to planning, but deficient in interpretability
  (2) Explicit Design: Some crucial components are still missing

New Paradigm

An end-to-end framework

- **Joint optimization**
- **Five crucial AD tasks**
- **Efficient task communication**
- Towards safe-planning



Method: UniAD Framework

Method

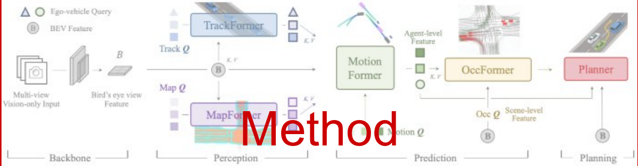
Q: Which tasks should be incorporated in an AD pipeline?

Track agent behavior — Tracking | Motion — Interact with all agents
Follow map guidance — Mapping | Occupancy — Find free space
Perception | Prediction → Safe Planning

A: We investigate five tasks in perception and prediction to facilitate planning.

- **Agent-wise**: Detection/Tracking, Motion Forecasting
- **Whole-scene**: Mapping, Occupancy Prediction

Q: How to coordinate various tasks to assist planning?



Unified Query

- Track Q : one query for one agent
- Map Q : one query for one map element
- Motion Q : one query for one trajectory
- Occ Q : one query for one BEV grid

A: Each tasks module is a transformer-based structure, where *attention* mechanisms model complex relations in the scene, and *queries* interact between tasks modules and transmit upstream knowledge to final planner.

Ps. Please see details in our paper and supp.

Q: Any special designs?

A:

- **Ego-vehicle query** to specifically model the self-car
- **Planning optimization** with future occupancy
- **Non-linear optimization** for upstream error recovery

Q: How to train such a network?

A:

- **Two phases training**. Perception + all modules end-to-end
- **Shared matching**. Matching results of tracking reused in motion and occ

Results

Ablation Study of Task Coordination

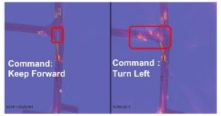
ID	Modules: Track	Map	Motion	Occ.	Plan	Tracking: AMOTA↑	AMOTP↓	IDS↓	Mapping: IoU-lane↑	IoU-road↑	Motion Forecasting: minADE↓	minFDE↓	MR↓	Occupancy Prediction: IoU-n↑	IoU-f↑	VPQ-n↑	VPQ-f↑	Planning: avg.L2↓	avg.Col↓
0*	✓	✓	✓	✓	✓	0.356	1.328	893	0.302	0.675	0.858	1.270	0.186	55.9	34.6	47.8	26.4	1.154	0.941
1	✓					0.348	1.333	791	-	-	-	-	-	-	-	-	-	-	-
2		✓				-	-	-	**0.305**	0.674	-	-	-	-	-	-	-	-	-
3	✓	✓				0.355	1.336	785	0.301	0.671	-	-	-	-	-	-	-	-	-
4			✓			-	-	-	-	-	0.815	1.224	0.182	-	-	-	-	-	-
5	✓		✓			0.360	1.350	919	-	-	0.751	1.109	0.162	-	-	-	-	-	-
6	✓	✓	✓			0.354	1.339	820	0.303	0.672	0.736(-9.7%)	1.066(-12.9%)	0.158	-	-	-	-	-	-
7				✓		-	-	-	-	-	-	-	-	60.5	37.0	52.4	29.8	-	-
8	✓			✓		0.360	**1.322**	809	-	-	-	-	-	62.1	38.4	52.2	32.1	-	-
9	✓	✓	✓	✓		0.359	1.359	1057	0.304	**0.675**	**0.710**(-3.5%)	**1.005**(-5.8%)	**0.146**	**62.3**	39.4	**53.1**	32.2	-	-
10					✓	-	-	-	-	-	-	-	-	-	-	-	-	1.131	0.773
11	✓	✓	✓		✓	**0.366**	1.337	889	0.303	0.672	0.741	1.077	0.157	-	-	-	-	1.014	0.717
12	✓	✓	✓	✓	✓	0.358	1.334	**641**	0.302	0.672	0.728	1.054	0.154	**62.3**	**39.5**	52.8	**32.3**	**1.004**	**0.430**

- L4-6: Tracking & Mapping -> (+) Motion
- L7-9: Motion <-> Occupancy
- L10-12: Motion & Occupancy & All -> Planning

Ps. Please see more results /ablations in our paper.

Attention Map in Planner

- The Planner attends to different areas given various high-level commands.

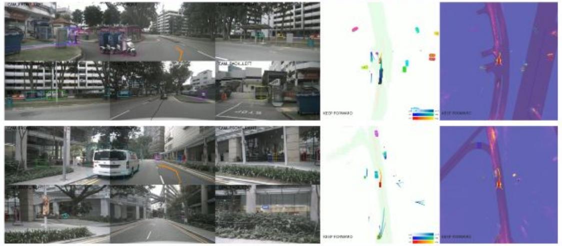


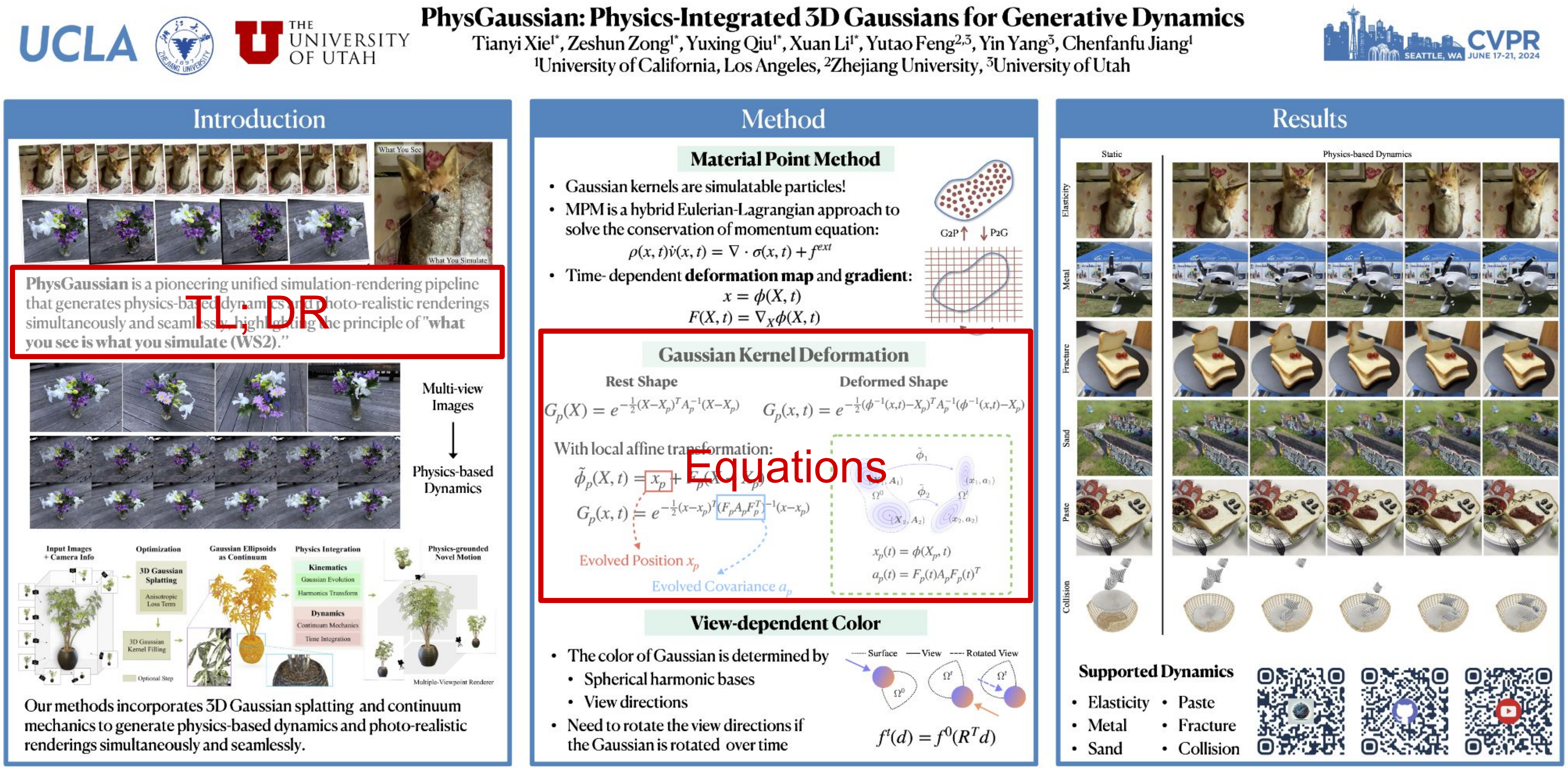
SOTA on nuScenes

- Lowest L2 error and collision rate, even outperforms LiDAR-based methods.

Method	L2(m)↓ 1s	2s	3s	Avg.	Col. Rate(%)↓ 1s	2s	3s	Avg.
NMP† [101]	-	-	2.31	-	-	-	1.92	-
SA-NMP† [101]	-	-	2.05	-	-	-	1.59	-
FF† [37]	0.55	1.20	2.54	1.43	0.06	0.17	1.07	0.43
EO† [47]	0.67	1.36	2.78	1.60	0.04	0.09	0.88	0.33
ST-P3 [38]	1.33	2.11	2.90	2.11	0.23	0.62	1.27	0.71
UniAD	**0.48**	**0.96**	**1.65**	**1.03**	**0.05**	**0.17**	**0.71**	**0.31**

Visualizations of Complex Scenes





PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics

Tianyi Xie[1*], Zeshun Zong[1*], Yuxing Qiu[1*], Xuan Li[1*], Yutao Feng[2,3], Yin Yang[3], Chenfanfu Jiang[1]

[1]University of California, Los Angeles, [2]Zhejiang University, [3]University of Utah

CVPR SEATTLE, WA JUNE 17-21, 2024

Introduction

What You See

What You Simulate

PhysGaussian is a pioneering unified simulation-rendering pipeline that generates physics-based dynamics and photo-realistic renderings simultaneously and seamlessly, highlighting the principle of "**what you see is what you simulate (WS2)**."

TL; DR

Multi-view Images → Physics-based Dynamics

Input Images + Camera Info · Optimization: 3D Gaussian Splatting, Anisotropic Loss Term, 3D Gaussian Kernel Filling, Optional Step · Gaussian Ellipsoids as Continuum · Physics Integration: Kinematics (Gaussian Evolution, Harmonics Transform), Dynamics (Continuum Mechanics, Time Integration) · Physics-grounded Novel Motion · Multiple-Viewpoint Renderer

Our methods incorporates 3D Gaussian splatting and continuum mechanics to generate physics-based dynamics and photo-realistic renderings simultaneously and seamlessly.

Method

Material Point Method

- Gaussian kernels are simulatable particles!
- MPM is a hybrid Eulerian-Lagrangian approach to solve the conservation of momentum equation:

$$\rho(x,t)\dot{v}(x,t) = \nabla \cdot \sigma(x,t) + f^{ext}$$

- Time- dependent **deformation map** and **gradient**:

$$x = \phi(X,t)$$
$$F(X,t) = \nabla_X \phi(X,t)$$

G2P ↑ ↓ P2G

Gaussian Kernel Deformation

Rest Shape

$$G_p(X) = e^{-\frac{1}{2}(X-X_p)^T A_p^{-1}(X-X_p)}$$

Deformed Shape

$$G_p(x,t) = e^{-\frac{1}{2}(\phi^{-1}(x,t)-X_p)^T A_p^{-1}(\phi^{-1}(x,t)-X_p)}$$

With local affine transformation:

$$\tilde{\phi}_p(X,t) = \boxed{x_p} + F_p(X - X_p)$$
$$G_p(x,t) = e^{-\frac{1}{2}(x-x_p)^T \boxed{(F_pA_pF_p^T)}^{-1}(x-x_p)}$$

Evolved Position x_p

Evolved Covariance a_p

$\tilde{\phi}_1$, $\tilde{\phi}_2$, Ω^0, Ω^t, (X_1, A_1), (x_1, a_1), (X_2, A_2), (x_2, a_2)

$$x_p(t) = \phi(X_p, t)$$
$$a_p(t) = F_p(t)A_pF_p(t)^T$$

Equations

View-dependent Color

- The color of Gaussian is determined by
  - Spherical harmonic bases
  - View directions
- Need to rotate the view directions if the Gaussian is rotated over time

Surface — View — Rotated View

$$f^t(d) = f^0(R^Td)$$

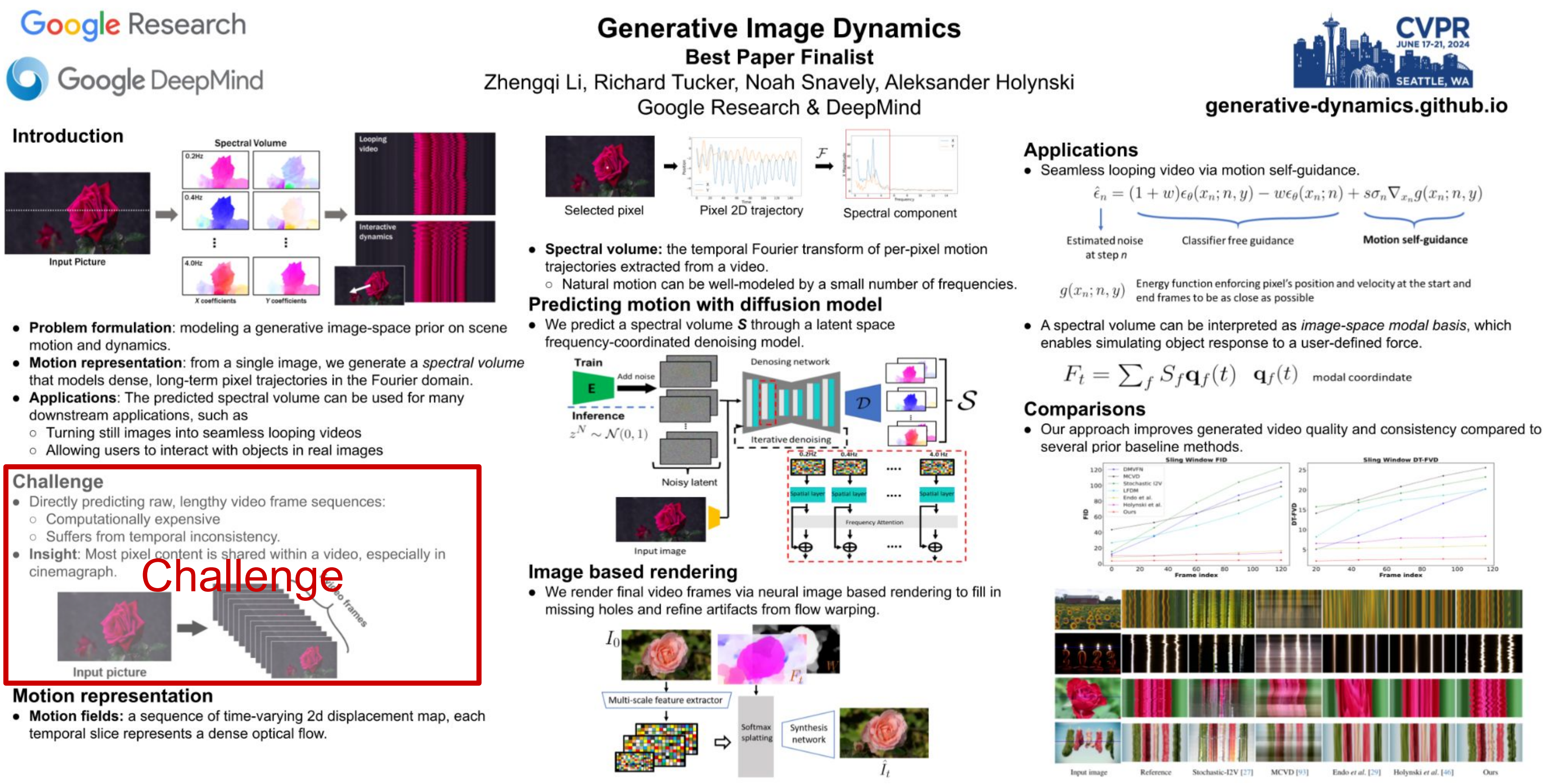
Results

Static · Physics-based Dynamics

Elasticity · Metal · Fracture · Sand · Paste · Collision

Supported Dynamics

- Elasticity
- Metal
- Sand
- Paste
- Fracture
- Collision



Google Research

Google DeepMind

Generative Image Dynamics

Best Paper Finalist

Zhengqi Li, Richard Tucker, Noah Snavely, Aleksander Holynski

Google Research & DeepMind

CVPR JUNE 17-21, 2024 SEATTLE, WA

generative-dynamics.github.io

Introduction

Input Picture

- **Problem formulation**: modeling a generative image-space prior on scene motion and dynamics.
- **Motion representation**: from a single image, we generate a *spectral volume* that models dense, long-term pixel trajectories in the Fourier domain.
- **Applications**: The predicted spectral volume can be used for many downstream applications, such as
  - Turning still images into seamless looping videos
  - Allowing users to interact with objects in real images

Challenge

- Directly predicting raw, lengthy video frame sequences:
  - Computationally expensive
  - Suffers from temporal inconsistency.
- **Insight**: Most pixel content is shared within a video, especially in cinemagraph.

Challenge

Input picture

Motion representation

- **Motion fields:** a sequence of time-varying 2d displacement map, each temporal slice represents a dense optical flow.

Selected pixel → Pixel 2D trajectory → $\mathcal{F}$ Spectral component

- **Spectral volume:** the temporal Fourier transform of per-pixel motion trajectories extracted from a video.
  - Natural motion can be well-modeled by a small number of frequencies.

Predicting motion with diffusion model

- We predict a spectral volume $\boldsymbol{S}$ through a latent space frequency-coordinated denoising model.

Image based rendering

- We render final video frames via neural image based rendering to fill in missing holes and refine artifacts from flow warping.

Applications

- Seamless looping video via motion self-guidance.

$$\hat{\epsilon}_n = \underbrace{(1+w)\epsilon_\theta(x_n; n, y) - w\epsilon_\theta(x_n; n)}_{\text{Classifier free guidance}} + \underbrace{s\sigma_n \nabla_{x_n} g(x_n; n, y)}_{\text{Motion self-guidance}}$$

Estimated noise at step *n*

$g(x_n; n, y)$ Energy function enforcing pixel's position and velocity at the start and end frames to be as close as possible

- A spectral volume can be interpreted as *image-space modal basis*, which enables simulating object response to a user-defined force.

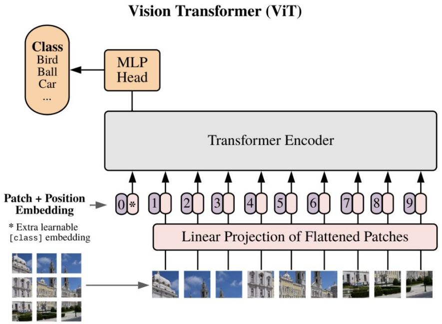
$$F_t = \sum_f S_f \mathbf{q}_f(t) \quad \mathbf{q}_f(t) \text{ modal coordindate}$$

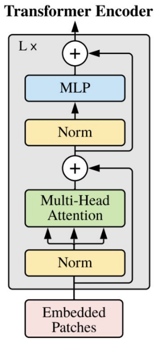
Comparisons

- Our approach improves generated video quality and consistency compared to several prior baseline methods.

System Diagram Design

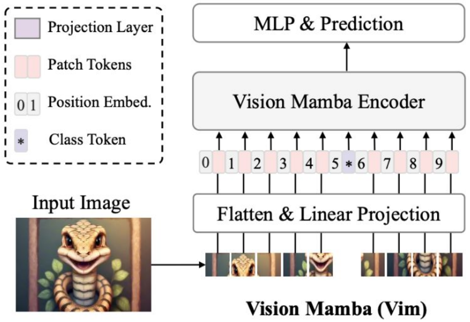
Vision Transformer

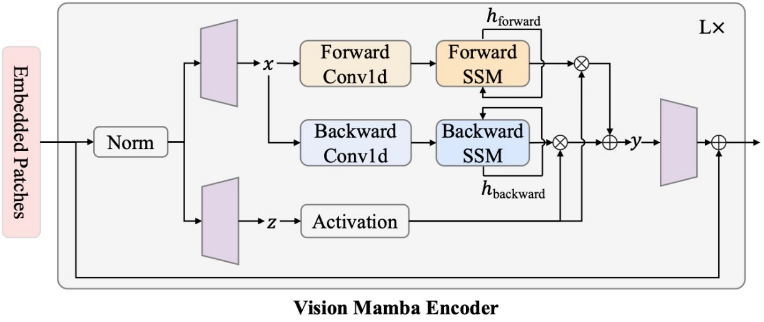




- Use draw.io or PowerPoint for poster creation
- Refer to top-conference papers for color schemes
- (use the color picker to extract colors)
- Avoid using text that is too small within figures

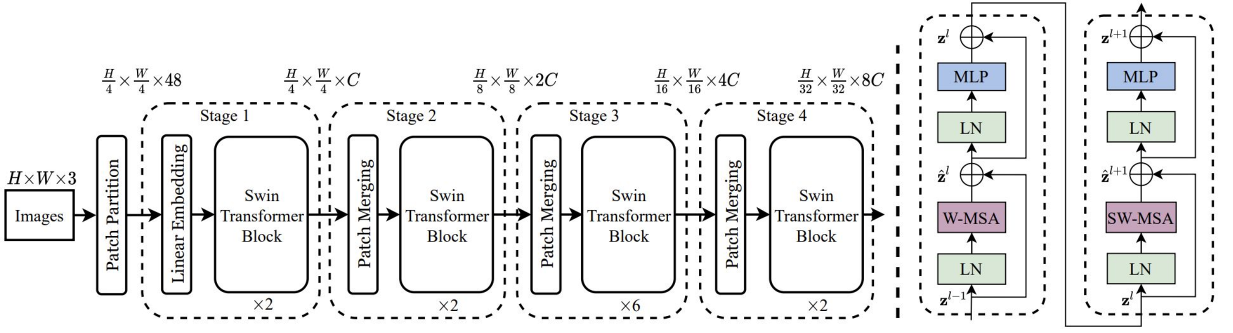
Vision Mamba





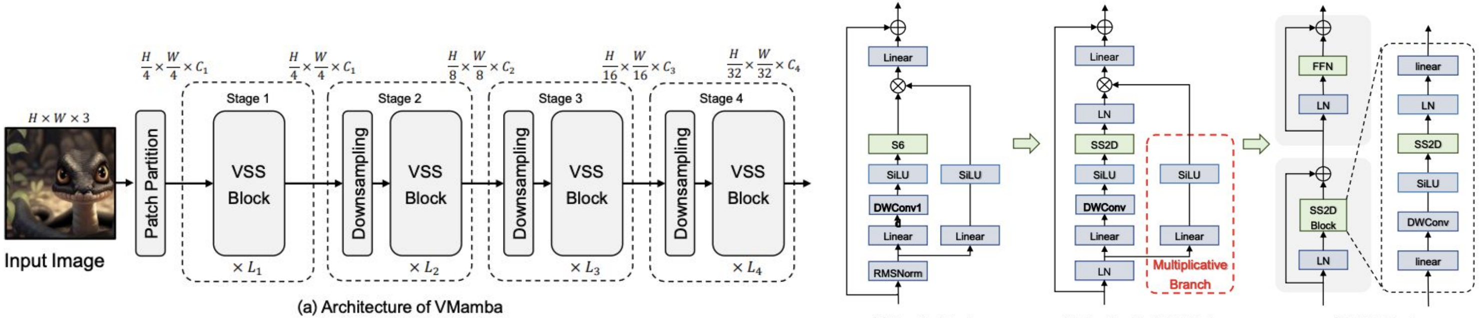
System Diagram Design

Swin Transformer



(a) Architecture
(b) Two Successive Swin Transformer Blocks

V-Mamba



(a) Architecture of VMamba
(b) Mamba Block
(c) The Vanilla VSS Block
(d) VSS Block

Experimental Tables

- Avoid enclosing tables with full borders
- Add ↑ or ↓ to indicate whether higher or lower values are better
- Highlight key results
  - First place: bold
  - Second place: underline
- Consider including standard deviations to show stability

	Methods	Adversarial Image Quality		
		SSIM ↑	PSNR ↑	LPIPS ↓
Church	PGAscent	0.37 ± 0.09	28.17 ± 0.22	0.73 ± 0.16
	Diff-Protect	0.39 ± 0.07	28.03 ± 0.12	0.67 ± 0.11
	AtkPDM (Ours)	0.75 ± 0.03	28.22 ± 0.10	0.26 ± 0.04
	AtkPDM^{+} (Ours)	$\mathbf{0.81} \pm 0.03$	$\mathbf{28.64} \pm 0.19$	$\mathbf{0.13} \pm 0.02$
Cat	PGAscent	0.48 ± 0.09	28.34 ± 0.18	0.65 ± 0.12
	Diff-Protect	0.33 ± 0.10	28.03 ± 0.15	0.80 ± 0.15
	AtkPDM (Ours)	0.71 ± 0.06	28.47 ± 0.18	0.29 ± 0.05
	AtkPDM^{+} (Ours)	$\mathbf{0.83} \pm 0.04$	$\mathbf{29.41} \pm 0.37$	$\mathbf{0.09} \pm 0.02$
Face	PGAscent	0.48 ± 0.05	$\mathbf{28.75} \pm 0.18$	0.64 ± 0.10
	Diff-Protect	0.25 ± 0.04	28.09 ± 0.20	0.91 ± 0.11
	AtkPDM (Ours)	0.56 ± 0.04	28.01 ± 0.22	0.36 ± 0.04
	AtkPDM^{+} (Ours)	$\mathbf{0.81} \pm 0.04$	28.39 ± 0.20	$\mathbf{0.12} \pm 0.03$

Ablation Study

DON'Ts

Method	mAP
[Snorlax et al. 2018]*	25.0
[Bulbasaur et al. 2019]*†	29.8
[Psyduck et al. 2020] †	32.1
Ours	35.5

DOs

Method	External data?	Finetuned?	mAP
[Snorlax et al. 2018]	✓	-	25.0
[Bulbasaur et al. 2019]	✓	✓	29.8
[Psyduck et al. 2020]	-	✓	32.1
Ours	-	-	35.5

Reference

- https://github.com/zhoubolei/bolei_awesome_posters
- https://github.com/jbhuang0604/awesome-tips#writing

Thank you